

MAE 140 Winter 12 Homework #1

TR & T, 1.14, 1.22, 1.24, 2.2, 2.5, 2.11, 2.17, 2.23, 2.28, 2.38

Due: 01/17/201

1-14 An incandescent lamp absorbs 100 W when connected to a 120 V source. A fluorescent lamp producing the same amount of light absorbs 12 W when connected to the same source. How much cheaper is it to operate the fluorescent bulb rather than the incandescent bulb over 1000 hours when electricity costs 7.2 cents/kWh?

1-22 Figure P1-22 shows an electric circuit with a voltage and current variable assigned to each of the six devices. The device voltages and currents are observed to be

	v (V)	i (A)
Device 1	15	-1
Device 2	5	1
Device 3	10	2
Device 4	-10	-1
Device 5	20	-3
Device 6	20	2

Find the power associated with each device and state whether the device is absorbing or delivering power. Use the power balance to check your work.

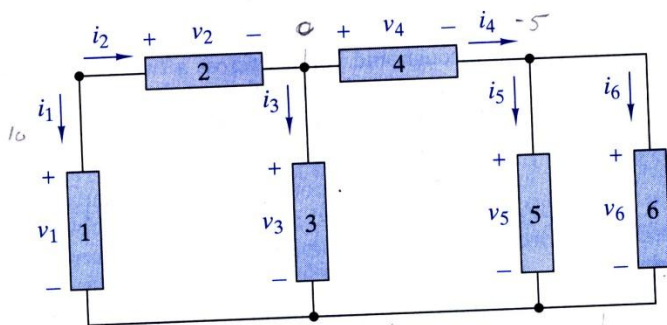


FIGURE P1-22

1-24 Suppose that in Figure P1-22 a ground is connected to the minus (-) side of device 6 and another to the junction of devices 2, 3 and 4. Further, assume that the voltage v_4 is 5 V and v_1 is 10 V. What are the voltages v_2 , v_3 , v_5 , and v_6 ?

2-2 A 6.2-k Ω resistor dissipates 12 mW. Find the current through the resistor.

2-5 In Figure P2-5 find R_x and the power delivered to the resistor.

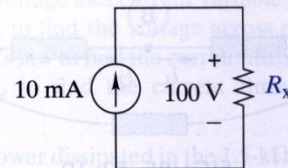


FIGURE P2-5

2-11 For the circuit in Figure P2-11:

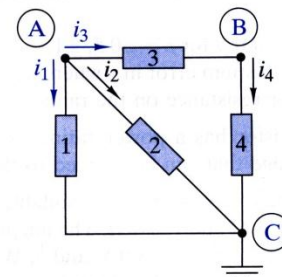


FIGURE P2-11

- Identify the nodes and at least two loops.
- Identify any elements connected in series or in parallel.
- Write KCL and KVL connection equations for the circuit.

2-17 In Figure P2-17 $i_2 = 10 \text{ mA}$, $i_3 = -15 \text{ mA}$, and $i_4 = 5 \text{ mA}$. Find i_1 and i_5 .

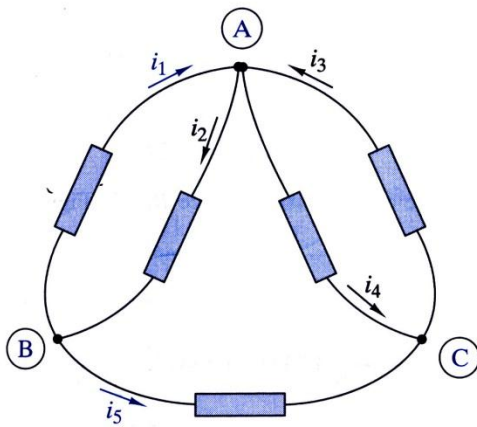


FIGURE P2-17

2-28 Find the equivalent resistance R_{EQ} in Figure P2-28.

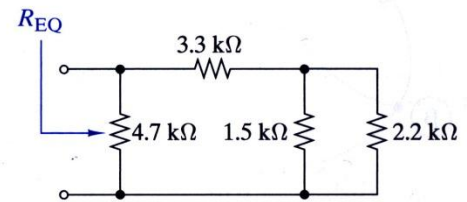


FIGURE P2-28

2-23 Find the power dissipated in the $1.5\text{-k}\Omega$ resistor in Figure P2-23.

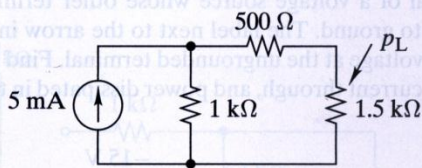


FIGURE P2-23

2-38 Select the value of R in Figure P2-38 so that $R_{AB} = R_L$.

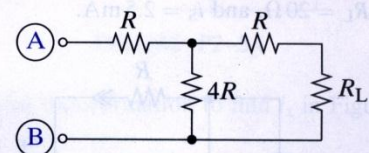


FIGURE P2-38